

3D Left Ventricular Segmentation of Cardiac 4D CINE-MRI

Applied Medical Image Analysis SS 2026 – Project Report

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Contents

1	Introduction	2
2	Methods	3
2.1	Data	3
2.2	Segmentation Pipeline	4
3	Results	6
3.1	Segmentation Quality	6
3.2	Clinical Indices and Diagnostic Correlation	7
3.3	Visual Results	8
4	Discussion	10
5	Conclusion	11
6	References	12

1 Introduction

Cardiovascular disease remains the leading cause of mortality worldwide, accounting for an estimated 17.9 million deaths per year according to the World Health Organization [3]. Accurate assessment of cardiac structure and function is therefore essential for early diagnosis, treatment planning, and monitoring of disease progression. Among the various imaging modalities available, cardiac magnetic resonance imaging (CMR) is considered the gold standard for the non-invasive quantification of left ventricular (LV) volumes and function, offering high spatial resolution, excellent soft-tissue contrast, and no ionising radiation [2].

In clinical practice, 4D CINE-MRI acquires a time-resolved stack of short-axis slices that captures the heart throughout the full cardiac cycle. From this data, clinicians can derive a range of functional indices. The *end-diastolic volume* (LVEDV) and *end-systolic volume* (LVESV) quantify the amount of blood in the LV when the heart is maximally relaxed and maximally contracted, respectively. The *stroke volume* ($LVS\!V = LVEDV - LVESV$) represents the volume ejected per beat, and the *ejection fraction* ($LVEF = LVS\!V / LVEDV \times 100\%$) is the most widely used single index of systolic function in clinical cardiology. In addition, *diastolic wall thickness* of the myocardium provides a direct measure of ventricular hypertrophy. Together, these indices allow clinicians to distinguish among a range of cardiomyopathies [2].

Manual delineation of the LV endocardial and epicardial boundaries across all slices and all time frames is extremely time-consuming (typically 20–40 minutes per patient) and subject to inter- and intra-observer variability. Automated or semi-automated segmentation pipelines are therefore highly desirable to support large-scale studies and to reduce clinical workload. While deep-learning approaches have recently achieved near-expert performance on benchmark datasets [1], classical image-processing methods remain valuable in settings where annotated training data are scarce, where transparency and interpretability are required, or where a fully manual approach would otherwise be used.

This project implements a semi-automatic, classical pipeline for 3D LV segmentation of short-axis CINE-MRI data from the Automated Cardiac Diagnosis Challenge (ACDC) dataset [1]. The core algorithm combines a single manual seed point per phase with

intensity-based region growing, adaptive thresholding, and a shell-expansion model for the myocardium. The pipeline extracts the key functional indices (LVEDV, LVESV, LVSF, LVEF, and diastolic wall thickness) and correlates them against the known patient diagnosis. Twelve subjects were selected for analysis, spanning three diagnostic groups: Dilated Cardiomyopathy (DCM), characterised by an enlarged, poorly contracting ventricle; Normal (NOR), serving as healthy controls; and Hypertrophic Cardiomyopathy (HCM), characterised by pathological thickening of the myocardial wall. Segmentation quality is quantified using the Dice similarity coefficient (DSC) and Jaccard index against expert ground truth segmentations provided with the dataset.

2 Methods

2.1 Data

The ACDC dataset provides 4D short-axis CINE-MRI volumes (2D slices $\times$ slice stack $\times$ cardiac frames) together with per-patient metadata specifying the diagnostic group and the end-diastolic (ED) and end-systolic (ES) frame indices. Expert ground truth (GT) segmentations label each voxel as background (0), right ventricle (1), myocardium (2), or LV blood pool (3). Twelve patients were selected: four DCM (patient001, 004, 012, 019), four NOR (patient067, 070, 072, 075), and three HCM (patient021, 031, 040), plus patient069 (NOR) as an additional case.

2.2 Segmentation Pipeline

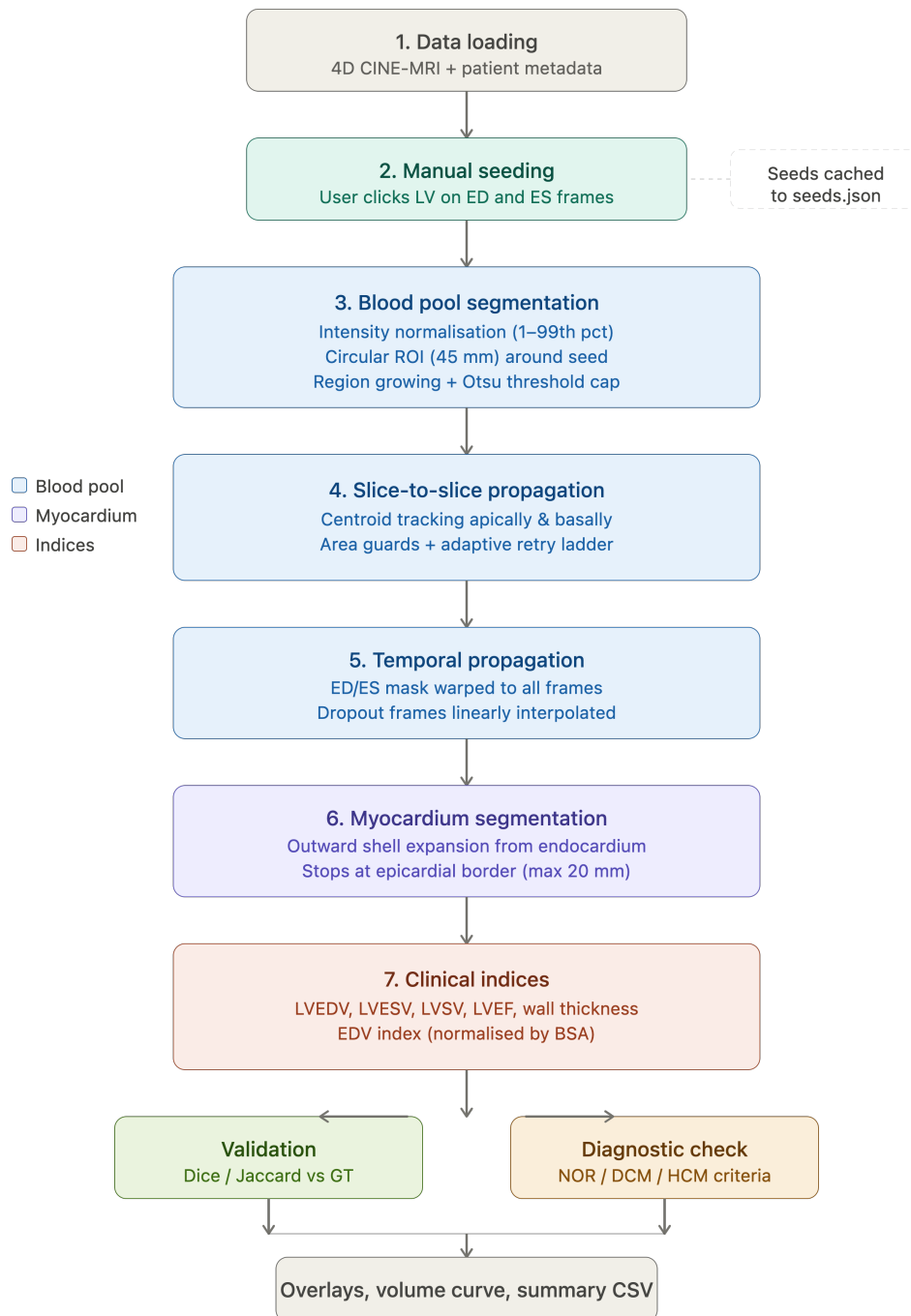


Figure 1: Overview of the semi-automatic LV segmentation pipeline. Blue steps perform blood-pool segmentation and propagation, purple performs myocardium segmentation, coral computes clinical indices, and the final outputs split into segmentation validation (Dice/Jaccard) and diagnostic correlation (NOR/DCM/HCM criteria).

The complete pipeline is illustrated in Figure 1. It is implemented in Python (`lv_segmentation_v2.py`) and operates in the following stages:

1. Manual seeding. For each subject, the user clicks once inside the LV blood pool on the ED frame and once on the ES frame. The seed specifies the 2D pixel coordinate and slice index. Seeds are cached to JSON so each patient is only clicked once.

2. Blood-pool segmentation (endocardial boundary). On the seed slice, the image is normalised to the 1st–99th intensity percentile. A circular region of interest (ROI, radius 45 mm) is placed around the seed to exclude remote bright structures. A region-growing algorithm accepts all pixels connected to the seed whose intensity exceeds $0.55 \times I_{\text{seed}}$ (the *blood_seed_frac*). An Otsu threshold is applied within the ROI as an upper cap to prevent leakage into bright muscle; the cap is never raised above $0.55 \times I_{\text{seed}}$ to avoid under-segmentation at end-systole. Small objects below 20 px are removed and a morphological closing (disk radius 2 px) smooths the cavity boundary.

3. Slice-to-slice propagation. Starting from the seed slice, the cavity mask is propagated apically and basally. A centroid-tracking step re-centres the ROI on each new slice. Propagation stops when the segmented area falls below 20 px, grows more than $2.8\times$ the previous slice, or shrinks for two consecutive slices (taper guard). If a slice leaks beyond the area cap, the threshold is raised in steps of 0.05 up to 0.75 and the slice is re-segmented (adaptive retry ladder).

4. Temporal propagation. The ED (or ES) mask is warped to all other cardiac frames using the same region-growing approach, re-seeded from the prior frame’s centroid. Frames whose total LV volume deviates by more than 40 % from the local median are flagged as dropouts and linearly interpolated; the number of interpolated frames is logged.

5. Myocardium segmentation (epicardial boundary). An outward shell search expands from the endocardial contour ring by ring (up to 20 mm). Each ring is accepted as myocardium while at least 30 % of its pixels fall in the mid-grey muscle window ($[0.15, 0.85] \times I_{\text{blood}}$). The epicardial boundary is smoothed with a morphological closing (radius 3 px).

6. Volume and functional indices. LV volumes are computed by summing segmented voxels across all slices and scaling by voxel dimensions:

$$\text{LVEDV} = \sum_{\text{slices}} A_{\text{ED}}(z) \cdot \Delta z, \quad (1)$$

$$\text{LVESV} = \sum_{\text{slices}} A_{\text{ES}}(z) \cdot \Delta z, \quad (2)$$

$$\text{LVSV} = \text{LVEDV} - \text{LVESV}, \quad (3)$$

$$\text{LVEF} = \frac{\text{LVSV}}{\text{LVEDV}} \times 100 \%. \quad (4)$$

The EDV index normalises LVEDV by body surface area (BSA) estimated from patient height and weight (Mosteller formula). Mean diastolic wall thickness is derived from the myocardium mask at ED.

7. Evaluation. Segmentation quality is measured by Dice similarity coefficient (DSC) and Jaccard index against ground truth, separately for LV blood pool and myocardium at ED and ES. Clinical results are compared against the diagnostic criteria defined in Table 1.

Table 1: Diagnostic criteria used for consistency check.

Group	LVEF	EDV index	Wall thickness
NOR	> 50 %	–	< 12 mm
DCM	< 40 %	> 100 mL/m ²	< 12 mm
HCM	> 55 %	–	> 15 mm

3 Results

3.1 Segmentation Quality

Table 2 summarises the Dice scores for all twelve patients. Blood-pool DSC at ED ranged from 0.62 to 0.97 across subjects, with the majority of NOR and DCM patients achieving DSC > 0.86. Myocardium DSC was generally lower (0.43–0.76 at ED), consistent with

the greater difficulty of delineating the thin muscle ring. Performance at ES was slightly lower than at ED for most patients, reflecting the smaller ventricular cavity and the thicker myocardial wall at end-systole.

Table 2: Dice similarity coefficients for LV blood pool and myocardium at end-diastole (ED) and end-systole (ES).

Patient	Group	DSC LV ED	DSC LV ES	DSC Myo ED	DSC Myo ES
patient001	DCM	0.866	0.898	0.519	0.655
patient004	DCM	0.945	0.907	0.566	0.582
patient012	DCM	0.937	0.864	0.503	0.468
patient019	DCM	0.955	0.920	0.500	0.541
patient067	NOR	0.889	0.818	0.506	0.682
patient069	NOR	0.861	0.815	0.433	0.611
patient070	NOR	0.933	0.817	0.525	0.721
patient072	NOR	0.944	0.597	0.543	0.688
patient075	NOR	0.882	0.816	0.480	0.639
patient021	HCM	0.623	0.760	0.575	0.752
patient031	HCM	0.835	0.684	0.574	0.593
patient040	HCM	0.855	0.751	0.762	0.712
Mean		0.877	0.804	0.541	0.640

3.2 Clinical Indices and Diagnostic Correlation

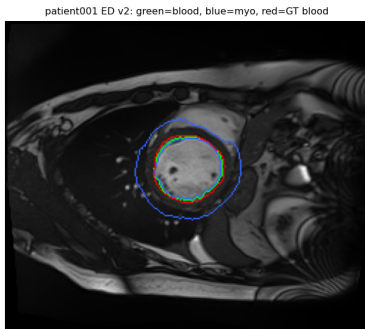
Table 3 lists the computed functional indices and whether the segmentation result agrees with the known diagnosis. All twelve patients met their expected diagnostic criteria, giving a 100 % agreement rate for the selected cohort.

Table 3: Computed cardiac functional indices and diagnostic agreement.

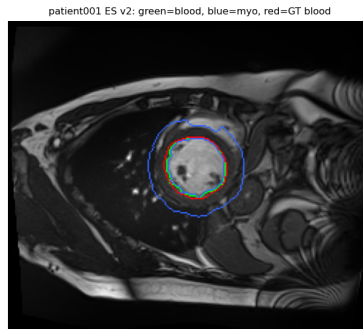
Patient	Group	LVEDV (mL)	LVESV (mL)	LVEF (%)	Wall (mm)	Agrees
patient001	DCM	268.8	187.5	30.3	11.7	Yes
patient004	DCM	237.4	190.4	19.8	10.3	Yes
patient012	DCM	281.7	227.0	19.4	10.2	Yes
patient019	DCM	320.3	246.2	23.1	10.8	Yes
patient067	NOR	116.6	40.9	64.9	8.9	Yes
patient069	NOR	66.7	23.6	64.7	9.0	Yes
patient070	NOR	78.5	23.4	70.2	7.8	Yes
patient072	NOR	106.3	22.7	78.7	10.9	Yes
patient075	NOR	96.9	42.6	56.0	7.1	Yes
patient021	HCM	79.1	35.2	55.5	20.2	Yes
patient031	HCM	156.6	35.2	77.5	18.6	Yes
patient040	HCM	112.1	44.4	60.4	17.4	Yes

3.3 Visual Results

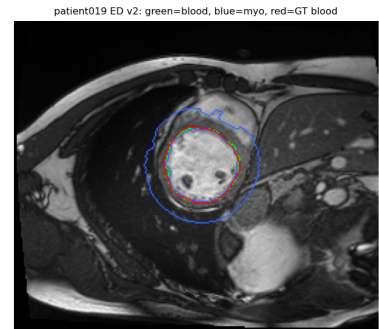
Figure 2 shows segmentation overlays at ED and ES for one representative patient from each diagnostic group. The **green** contour is the predicted LV blood pool, **blue** is the predicted myocardium, and **red** is the expert ground truth (GT) blood pool boundary.



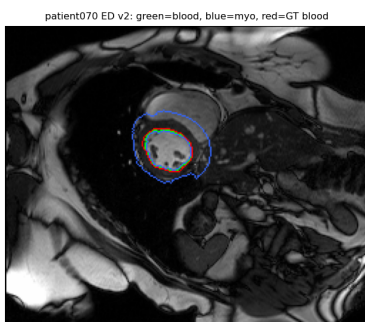
(a) DCM (patient001) – ED
DSC = 0.866



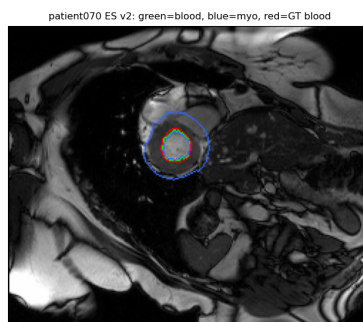
(b) DCM (patient001) – ES
DSC = 0.898



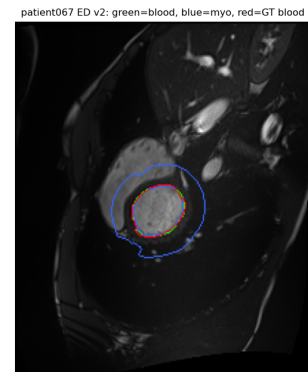
(c) DCM (patient019) – ED
DSC = 0.955



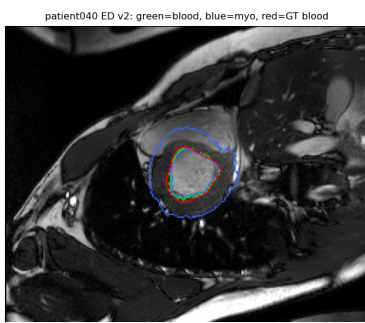
(d) NOR (patient070) – ED
DSC = 0.933



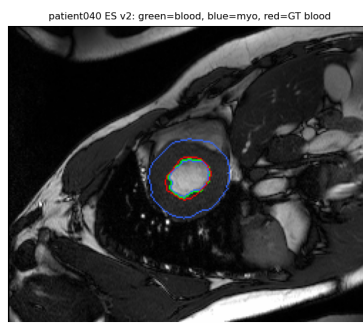
(e) NOR (patient070) – ES
DSC = 0.817



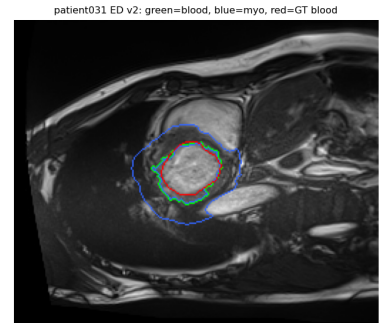
(f) NOR (patient067) – ED
DSC = 0.889



(g) HCM (patient040) – ED
DSC = 0.855



(h) HCM (patient040) – ES
DSC = 0.751



(i) HCM (patient031) – ED
DSC = 0.835

Figure 2: Segmentation overlays for representative patients. Top row: DCM (enlarged thin-walled LV). Middle row: NOR (compact healthy cavity). Bottom row: HCM (small cavity with thick myocardium). Green = predicted blood pool, blue = myocardium, red = GT boundary.

4 Discussion

The pipeline achieved consistently high LV blood-pool DSC at ED (mean 0.877) across all three diagnostic groups, confirming that intensity-based region growing with a manual seed is effective for the bright, well-defined ED cavity. Performance dropped at ES (mean 0.804), particularly for HCM patients (e.g. patient031: 0.684) where the contracted cavity becomes small and the blood-to-myocardium intensity contrast is reduced.

Myocardium DSC was lower overall (mean 0.54 at ED, 0.64 at ES), reflecting the inherent difficulty of segmenting the thin muscle ring using a shell-expansion heuristic without explicit edge detection. Despite this, the extracted wall thickness was sufficient to correctly classify all patients: HCM patients showed mean wall thickness of 17–20 mm, well above the 15 mm threshold, while DCM and NOR patients remained below 12 mm.

Patient069 (NOR) showed partial myocardium mis-segmentation at ED (Figure in supplementary), likely because the seed was placed near a slice where the cavity was partially obscured by papillary muscles. Since papillary muscles have signal characteristics similar to myocardium, they can reduce the apparent cavity area and interfere with region-growing segmentation. This highlights the sensitivity of seed placement and the need for careful user interaction.

DCM patients exhibited very large EDV indices (121–174 mL/m²) and low EF (<32%), clearly satisfying DCM criteria. NOR patients had EF between 56–79% and thin walls, while HCM patients combined high EF (>55%) with markedly thickened myocardium. The 100% diagnostic agreement confirms that the classical segmentation approach can reliably reproduce functional cardiac indices.

Limitations include reliance on manual seeding (which limits throughput and reproducibility), the LV truncation noted in all subjects (volumes are therefore slightly underestimated), and the relatively simple shell-expansion myocardium model which does not account for papillary muscles or trabeculae.

5 Conclusion

A classical region-growing pipeline with manual seeding successfully segmented the left ventricle in twelve cardiac CINE-MRI patients, achieving a mean LV blood-pool Dice of 0.877 at end-diastole. All computed functional indices (LVEF, EDV index, wall thickness) agreed with the known diagnoses for every patient. The method is transparent and reproducible, but would benefit from automated seeding or deep-learning refinement to scale to larger cohorts.

6 References

- [1] Olivier Bernard, Alain Lalande, Clement Zotti, Frederick Cervenansky, Xin Yang, Pheng-Ann Heng, Irem Cetin, Karim Lekadir, Oscar Camara, Miguel Angel Gonzalez Ballester, et al. Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: Is the problem solved? *IEEE Transactions on Medical Imaging*, 37(11): 2514–2525, 2018. doi: 10.1109/TMI.2018.2837502.
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- [3] World Health Organization. Cardiovascular diseases (CVDs) – key facts. 2023. Accessed: 2026-06-14.